UV-C RADIATION IMPACT ON THE SENSORY PROPERTIES OF RED WINES

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ABSTRACT

This study investigated the influence of UV-C radiation on the sensory properties of two red wine varieties, Dunaj (Slovakia) and Lasina Crna (Croatia). The wines were exposed to three doses of UV-C radiation and compared to untreated controls using instrumental color analysis (electronic eye) and sensory evaluation by sensory panel. UV-C treatment showed a clear varietal response. Dunaj wines showed slight improvements in color and aroma with minimal oxidation, while Lasina Crna showed a decrease in aromatic quality and color intensity. Instrumental colorimetric analysis confirmed strong varietal differentiation, with specific color tones associated with changes in sensory quality. The results indicate that UV-C treatment can be beneficial for certain varieties of wine.

KEYWORDS: *wine, UV-C treatment, electronic eye, color, sensory analysis*

MATERIALS AND METHODS

Two red wine varieties were tested: Dunaj (Slovakia) and Lasina Crna (Croatia). For each variety, one control sample and three samples treated with UV-C radiation were prepared. The irradiation times applied were 13.45, 6.37 and 2.57 minutes, chosen to avoid excessive ozone formation and phenol degradation observed in previous studies.UV-C treatment was performed using a germicidal quartz UV-C lamp (Philips UV6, 6 W) in a continuous flow system. The lamp was placed centrally, while the wine circulated around it, creating a liquid layer approximately 5 mm thick. For each variety, one control and three experimental variants were prepared as follows: LASINA CRNA: variant 0 represents the sample without irradiation, variant 1 (13.45 min of irradiation), variant 2 (6.37 min), variant 3 (2.57 min). Similarly DUNAJ: variant A represents the sample without irradiation, variant B (13.45 min.), variant C (6.37 min.), variant D (2.57 min.).The sensory evaluation was performed by a panel of five trained evaluators. The deviations from the control samples were rated on a scale from -5 to +5 for color, aroma and oxidation.Instrumental color analysis was performed using an IRIS VA 400 electronic eye system (Alpha M.O.S., France), equipped with a CMOS camera in a controlled light chamber. Before analysis, an automatic calibration was performed using a standardized colorimetric reference. Wine samples were analyzed, only color tones exceeding 1% of the representation were included. The results are expressed in RGB and CIEL*a*b* coordinates. Multivariate statistical analysis was applied, including principal component analysis (PCA) and correlation analysis.

RESULTS

UV-C radiation produced different effects depending on the wine variety. In the case of Dunaj wine, UV-C treatment resulted in a subjective improvement of color intensity and a slight enhancement of aroma, especially at moderate exposure times. Less oxidative changes were detected at higher doses, but did not significantly reduce the overall sensory quality. In contrast, Lasina crna wines showed a clear negative response to UV-C treatment. The treated samples showed a decrease in the intensity and freshness of the aroma, with a less vivid color expression. Sensory deterioration was observed at higher levels of irradiation.

Colorimetric analysis confirmed strong varietal differentiation. The Dunaj samples showed more uniform color spectrum dominated by darker tones such as Wenge and Beaver, while Lasina black showed more color complexity with dominant tones including Cordovan, Dark Chestnut and Café au lait. PCA revealed a clear separation of samples by variety, with the first principal component explaining 98.46% of the total variance. Correlation analysis showed that certain color tones (Cordovan, Dark Chestnut, Café au lait) were strongly negatively correlated with aroma scores, while Wenge and Beaver tones were positively associated with higher sensory quality.

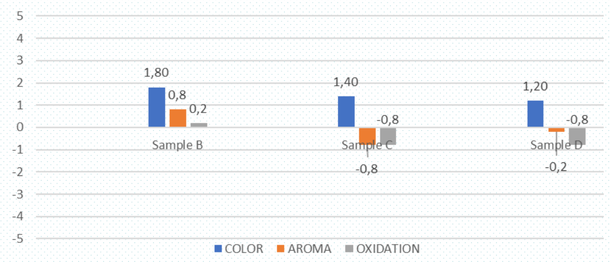


Figure 1: Dunaj variety: average deviations in the sensory scoring of samples B; C and D compared to the control sample A at sensory evaluation

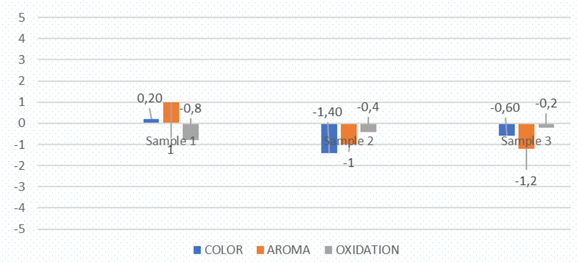
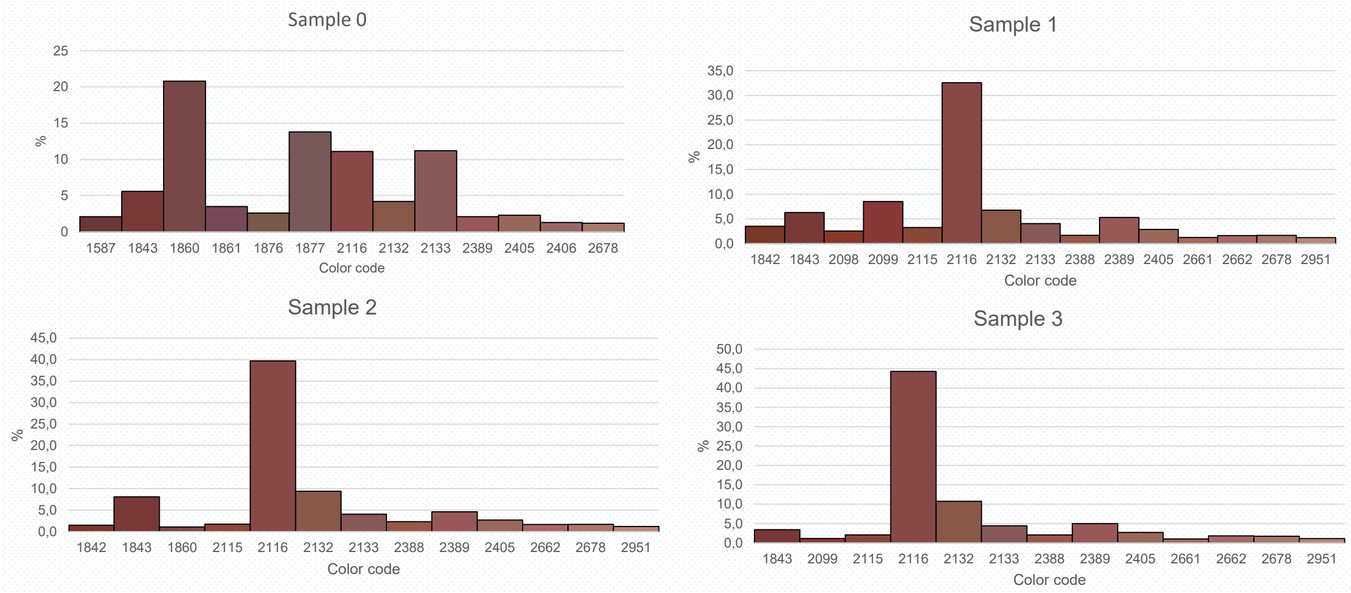
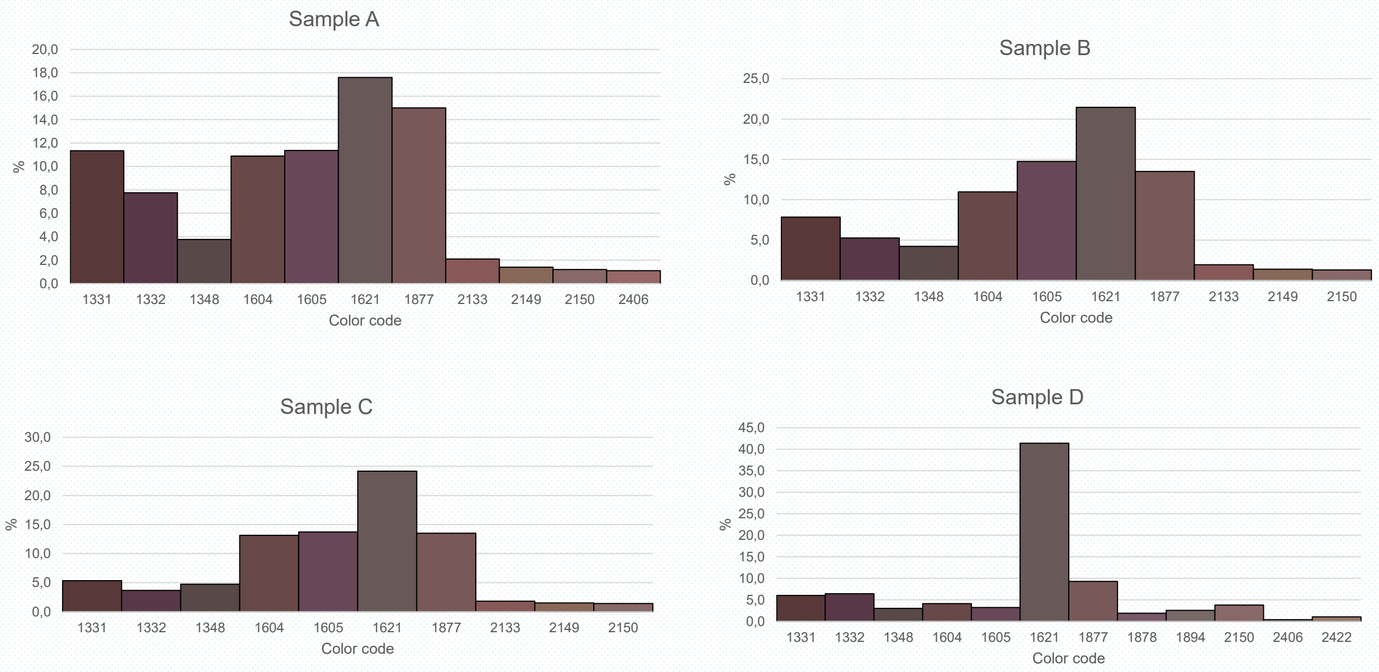


Figure 2: Lasina crna: average deviations in the sensory scoring of samples 1; 2 and 3 compared to the control sample 0:

Figure 3: Color spectra of wine samples treated with UV-C





CONCLUSION

UV-C irradiation did not significantly alter the overall color spectrum, but had a strong varietal influence on sensory quality. Dunaj wines showed good tolerance and potential sensory benefits from UV-C treatment, suggesting suitability for technological application. In contrast, Lasina Crna proved to be very sensitive, with noticeable aroma deterioration even at shorter exposure times. It seems the importance of varietal specificity when applying UV-C technology and suggest that sensitive wines require additional protective strategies. The combined use of sensory analysis and electronic eye colorimetry provides a powerful approach to assess the effects of UV-C irradiation on wine quality.

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